PLAIN-ENGLISH SYSTEM GUIDE

Quantara

How an automated research-to-live trading system turns market data into controlled MT5 decisions

Built by Wells

Prepared for friends, collaborators and curious non-technical readers

System basis: v2 · commit aa7632f7 · 12 August 2026

Read this first

Quantara is best understood as a disciplined automated decision system—not as a magic price predictor.

> **IN ONE SENTENCE** Quantara watches eight live markets, lets 74 specialist trading edges evaluate only the situations they were designed for, sizes approved opportunities under portfolio and prop-account limits, and sends orders to MetaTrader 5 only after a separate execution process passes every safety check.

The system is vertically integrated: the same project contains the research workflow, model artifacts, portfolio construction, live market-data pipeline, signal engine, execution controls, recovery logic and audit trail. That matters because a good backtest is useful only if live trading can reproduce the same decision with information that genuinely existed at that moment.

This guide describes the code and sealed configuration on the current v2 branch after the 12 August 2026 recovery hotfix. It explains what Quantara is designed to do. It does not claim that today's still-running VPS session has already proved its speed or profitability; those operational results must be judged from the logs after the run.

Quantara at a glance

Measure	Current repository / live-manifest snapshot
Configured markets	10
Deployability-passed edges	109
Broker-available markets and edges	8 markets · 76 edges
Selected live portfolio	74 edges: 60 long and 14 short
Live market-data rhythm	Adaptive information bars targeting about 50 bars per market per day
Live processes	Dashboard/orchestrator + independent MT5 executor
Current account profile	$5,000 phase-one evaluation contract
Primary principle	If required evidence is missing or inconsistent, fail closed

> **IMPORTANT DISTINCTION** An edge is not a trade. It is one market + one trading idea + one direction. Most edges are inactive on most bars, and even an approved edge can still be blocked by freshness, spread, entry drift, broker rules, minimum lot or portfolio risk.

1. The big idea

Think of Quantara as an investment committee whose members have sharply separated jobs.

The sensors: MT5 quote ticks supply live bid and ask prices. The system records exactly where its market information came from.

The clockmaker: Ticks are grouped into adaptive information bars. Busy periods produce bars faster; quiet periods produce them more slowly.

The specialists: Each edge watches for one type of opportunity, such as a failed breakdown, a trend pullback or a volatility cascade.

The judge: A trained model estimates whether the current candidate resembles historically successful candidates for that exact edge.

The portfolio officer: The allocator decides which edges belong together and how much relative risk weight each should receive.

The trader: The executor checks the real market and account, calculates a broker-valid lot size and sends the order only when fully armed.

The auditor: Append-only logs, state files, release identity and the closed-trade ledger preserve what the system saw and why it acted or refused to act.

What problem it is solving

A trading idea can look good in research and still fail live for mundane reasons: a feature is calculated differently, a probability threshold changes, the broker clock is misread, a minimum lot is too risky, or a stale signal arrives after the price has moved. Quantara's architecture is built around closing those gaps between research, portfolio design and actual execution.

- Research and live scoring call the same edge runtime instead of maintaining two versions of the strategy.
- Every selected edge is sealed by schemas, artifacts, probability calibration, trade geometry and a deployability manifest.
- The dashboard creates decisions but cannot place orders; the executor places orders but cannot invent or rescore ideas.
- Risk is measured on broker-executable lots and includes open positions, pending orders and unmanaged account exposure.
- Operational failures are treated as reasons to pause or quarantine, not as permission to guess.

2. From a quote to a trade

The easiest way to understand the whole system is to follow one decision from left to right.

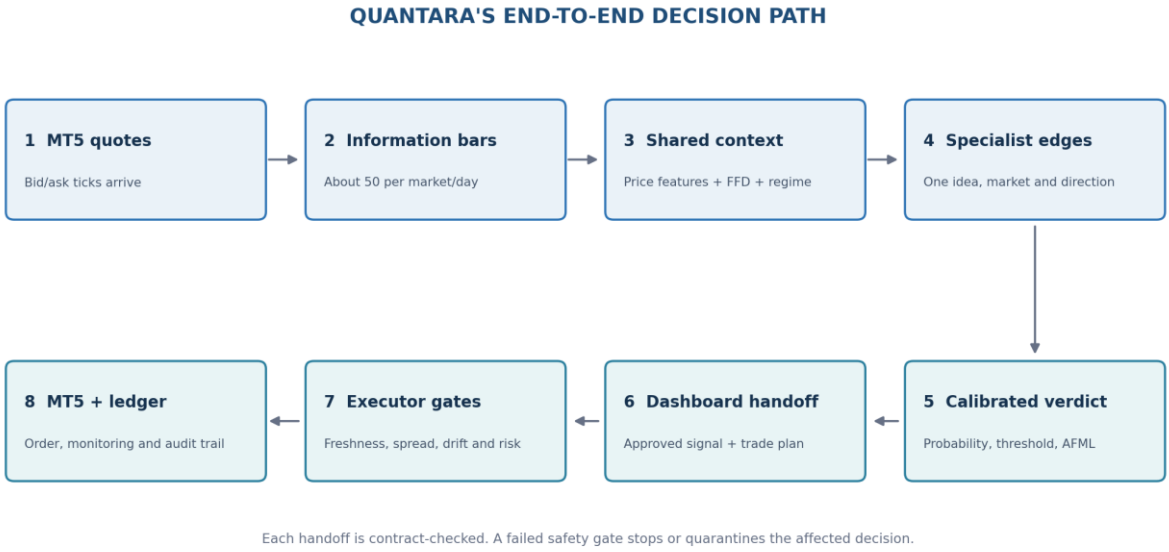


Figure 1. Quantara's end-to-end decision path.

1. MT5 supplies bid/ask quote ticks for each configured broker symbol.
2. The dashboard converts them into completed information bars using an adaptive number of ticks per bar.
3. Shared price features, fractionally differentiated price context and market-regime context are prepared from closed bars only.
4. Each relevant edge checks whether its own broad candidate event has occurred.
5. The edge's XGBoost classifier produces a raw probability, which is calibrated and compared with that edge's live threshold; payoff-aware AFML converts confidence into a sizing strength.
6. The dashboard emits an approved signal and an edge-specific execution plan. It still sends no order.
7. The executor revalidates identity, freshness, geometry, spread, entry drift, stops, account rules and portfolio risk using current broker information.
8. If every gate passes, the executor sends the MT5 order, records its metadata and later reconciles the position and close into the ledger.

3. The markets and the specialist portfolio

The repository is scalable to more markets, but the current broker and allocator determine what is actually active.

The asset registry contains ten markets. Hong Kong (HK) and USD/MXN are fully represented in research, but 33 edges attached to broker-unavailable symbols were skipped during active-universe ingestion. The current live portfolio therefore uses the following eight markets.

Market	Plain-English identity	Edges	Long / short	Relative weight
USOIL	WTI crude oil	5	2 / 3	25.29%
BTCUSD	Bitcoin / US dollar	14	5 / 9	21.25%
XAUUSD	Gold / US dollar	9	9 / 0	20.46%
NASDAQ	US technology index	10	8 / 2	14.27%
US500	S&P 500 index	9	9 / 0	8.05%
GBPJPY	Pound / yen FX cross	9	9 / 0	4.55%
DAX	German equity index	9	9 / 0	3.90%
JPN	Japanese equity index	9	9 / 0	2.22%

> **HOW TO READ THE WEIGHT COLUMN** These percentages are relative portfolio risk weights, not cash allocations. The actual risk of any trade is further reduced or capped by model confidence, AFML sizing, account throttles, open-risk limits, broker lot steps and the available drawdown buffer.

Of the 76 broker-available edges, the final portfolio selected 74. BTCUSD/trendpullbackcontinuation_long and USOIL/drawupreversalhazard_short remained available but were not included in the sealed live portfolio. The live manifest—not a folder's mere existence—is the final authority.

The live edge families in ordinary language

Edge family	Candidate thesis
Absorption	A downward push is rejected, suggesting buyers absorbed selling pressure.
Breakdown–retest failure	Support breaks, price retests it, then fails to reclaim it; a continuation-short setup.
Compression breakout	A relatively tight price range is followed by a first break beyond its structure.
Daily range completion fade	Price has stretched toward its expected daily range and may exhaust into a reversal.
Downside jump continuation	A negative one-bar move is unusually large relative to robust recent volatility.
Downside volatility cascade	Short-term variance accelerates and is dominated by negative returns.

Edge family	Candidate thesis
Drawdown hazard continuation	A drawdown state historically appears more likely to make another trough before recovering.
Range structure	Price is near a prior range extreme while a past-only structural bias points one way.
Reversal	Price is unusually stretched or oversold and may revert toward fair value.
Sign-entropy collapse	Return directions become less random, downside-dominated and conditionally persistent.
Time regime	Past-only time and market-state tendencies create a directional bias.
Trend	A directional impulse is supported by a slower, memory-preserving trend context.
Trend-pullback continuation	A medium-term trend pauses or pulls back, then begins to reclaim direction.
Volatility carry	The prevailing volatility state supports continuation rather than immediate reversal.
Volatility regime	A learned market-state pattern makes one side more attractive than usual.

The family name describes the situation that opens the question. It is not the final answer. Candidate rules are deliberately broad; the edge's model uses many contextual features to decide whether the present example is strong enough.

4. How Quantara reads the market

Quantara measures market activity in information, not only in minutes.

Quote ticks become adaptive bars

A quote tick is an update to the broker's bid or ask price. Quantara groups a market-specific number of quote ticks into an OHLC bar. The policy targets about 50 completed bars per market per day. Because the threshold adapts from completed historical days, a fast market usually closes bars sooner and a quiet market more slowly.

- The adaptive threshold uses a 20-day exponentially weighted average of completed daily tick counts.
- The current incomplete day is excluded, so tomorrow's threshold cannot borrow information from tomorrow.
- The threshold is frozen when a bar starts; it does not move halfway through that bar.
- The legacy field Proxy_Value means tick count only. It is not traded volume, dollar volume or liquidity.
- Bars reconstructed during startup gap healing restore memory but are marked non-executable, so history repair cannot manufacture a live trade.

> **WHY THIS MATTERS** A five-minute bar always lasts five minutes, whether the market barely moves or erupts. A quote-tick bar instead waits for a comparable amount of market updating, giving the research and live system a more consistent unit of information.

Shared market context

Every edge begins from the same closed-bar history for its asset. That shared layer supplies ordinary price statistics—returns, ranges, candles, moving averages, momentum and volatility—plus two important transformations.

Fractional differentiation (FFD): A mathematical way to reduce non-stationary drift while retaining more long-memory information than ordinary differencing. Its parameter is estimated in research, frozen and reused live; it is not re-estimated opportunistically.

Hidden Markov Model (HMM) regime: A four-state, price-only market-regime model. It uses closed-price and FFD context such as volatility, absolute return, candle body/range and close location. Walk-forward fitting and live continuation use past information only.

An edge can use these shared inputs, but it still owns its own event logic and selected model features. The dashboard is explicitly forbidden from hand-building a missing edge feature because doing so would create a second, unverified version of the strategy.

5. How an edge earns the right to go live

A promising idea is only the beginning. Quantara requires evidence, reproducibility and sealed handoffs.

The research sequence in plain English

1. **Define the event.** Specify one market, one family and one direction. The event generator must use closed, past-only information and must not contain labels, probabilities or trade decisions.
2. **Create honest outcomes.** For each candidate, define the entry timing, profit target, stop and maximum holding horizon. Labels ask whether the trade thesis succeeded under that frozen geometry.
3. **Select features and train.** Feature surgery reduces redundancy. An XGBoost classifier is tuned with Optuna and time-aware combinatorial purged cross-validation, including an embargo so overlapping future outcomes do not leak between training and test folds.
4. **Calibrate and test.** Out-of-fold raw probabilities are passed through sigmoid calibration. Threshold sweeps, runtime smoke tests and replay audits verify that the exact selected rows, probabilities and AFML sizes can be reproduced.
5. **Seal the edge.** The feature order, model input schema, model, calibrator, threshold artifact, trade geometry and runtime are recorded. Only a Step 09 PASS edge manifest can enter the allocator.

> **THE NON-NEGOTIABLE RULE** No edge is deployable unless the same edge runtime can reproduce its exact selected model-input rows live and pass the deployability audit.

Why each edge is side-specialized

A long edge and a short edge are separate packages. The model is not asked to choose direction; direction is already defined by the economic idea. The model's job is narrower: decide whether this particular candidate is likely to succeed for that side. Narrow responsibilities are easier to test and harder to accidentally change.

The probability decision

Each candidate moves through a fixed chain:

> **PROBABILITY FLOW** Raw model probability → sigmoid-calibrated probability → payoff-aware AFML sizing strength → edge-specific calibrated threshold decision.

Calibration: Makes a model score behave more like a usable probability. A model that says 70% should not merely rank candidates; across similar cases it should mean something close to 70%.

Threshold: The minimum calibrated probability required for that edge in the live portfolio. Current thresholds range from 0.31 to 0.58 and are not directly comparable across unrelated edges.

Payoff-aware AFML sizing: Converts probability above the edge's break-even point into a number between zero and one. A strategy with a 2:1 reward/risk ratio has a different economic break-even point from a 0.75:1 strategy, so confidence is interpreted in context.

Trade geometry: The immutable profit target, stop distance, reward/risk ratio and vertical holding horizon for that edge. Current selected edges use reward/risk ratios of 0.75, 1.0, 1.5 or 2.0 and horizons of 100, 250 or 1,000 information bars.

6. How 109 edges become one portfolio

A collection of individually good strategies can still be a bad portfolio if they all lose together.

The allocator's job

1. Ingest only edges whose deployability manifests say PASS.
2. Remove broker-unavailable markets and build a common candidate/trade bank.
3. Measure trade overlap, return correlation, common loss days, marginal contribution and standalone expectancy.
4. Compress thresholds into representative candidates instead of optimizing every tiny threshold difference.
5. Use hierarchical risk parity (HRP) to cluster similar return streams and allocate more balanced relative weights.
6. Convert research risk into real broker lot sizes, including minimum lot, lot step, tick value and currency conversion.
7. Replay rolling challenge starts under the active prop-account rules and select one coherent risk variant.
8. Run mandatory path-stress and Monte Carlo validation before exporting the live manifest.

Independence does not mean zero correlation

Quantara does not automatically delete two edges merely because their daily returns are correlated. It looks at whether they actually trigger the same trades, fail on the same days, add independent value and express the same economic idea. High correlation may justify a smaller shared cluster weight; it is not automatic proof of duplication.

What the stored stress evidence says

The sealed 21 July 2026 stress manifest reports 120 exact control paths under a 0.05R transaction-cost stress. It recorded zero hard rule breaches, 89 paths (74.2%) completing the target-and-minimum-days condition, and 31 paths ending the available history without completing it. A 1,500-sample month-cluster bootstrap at the same cost stress produced a median completion rate of about 74.2%, with a 95% interval of roughly 62.4% to 84.6%.

> **HOW TO INTERPRET THIS** This is evidence that the selected portfolio survived the tested historical paths and stress assumptions. It is not a guarantee of future profit, a guarantee against drawdown, or proof that live execution will match simulated fills.

7. The live system: dashboard and executor

Quantara deliberately separates deciding from trading.

Dashboard / orchestrator	Execution bot
Connects to MT5 market data	Connects to MT5 account and trade interface
Builds and repairs live information bars	Reads only dashboard-approved signals
Prepares exact shared feature snapshots	Revalidates schema, membership, age and geometry
Calls each selected edge's authoritative runtime	Checks spread, drift, market state, stops and lot rules
Applies calibrated thresholds and creates trade plans	Calculates broker-executable risk and may send the order
Cannot call order_send	Cannot build features, score models or choose thresholds

Dashboard startup and live loop

- Records a release identity, including the Git commit and critical runtime files.
- Validates the live manifest, account stage, broker symbols and production runtime contract.
- Builds a multi-symbol broker-clock consensus and converts broker server timestamps to UTC.
- Heals missing history, then marks repaired bars as memory-only rather than tradable.
- Queues newly closed asset bars, prepares an immutable feature generation and scores relevant edges.
- Writes runtime scores, the full audit funnel and approved-signal handoff files.

Executor decision gates

An approved model signal is only permission to consider a trade. Before an order is built, the executor checks the signal contract and the market as it exists now.

- Signal is approved, executable, fresh and not from a gap-healed bar.
- Edge, account stage, calibrated threshold, AFML strength and trade geometry match the live manifest.
- Broker-clock contract is fresh, locked and internally consistent; the normalized tick is neither stale nor genuinely in the future.
- Symbol is tradable, outside configured maintenance/weekend restrictions and not in a market-closed cooldown.
- Current spread and entry drift are within asset-specific limits.
- Stop and target respect broker minimum-distance rules.

- The proposed lot fits single-trade, edge, family, asset, total-open-risk and drawdown-buffer limits.
- Unknown position or pending-order risk causes a block rather than a guess.

> **NOT HIGH-FREQUENCY TRADING** Quantara is an event-driven intraday system, not a microsecond latency-arbitrage engine. The dashboard works from completed information bars, while the executor polls for approved signals on a one-second cadence and preserves a 1,000-second maximum signal-age contract.

How live order sending is armed

The tracked repository ships with safe defaults: dry-run mode, paper execution and order sending disabled. A private, untracked local settings file on the VPS can override those defaults. Real order sending still requires all of the following authorities to agree:

- local execution mode is live and ENABLE_ORDER_SEND is true
- local LIVE_MODE is REAL_LIVE and real-live export is allowed
- the live manifest is LIVE_APPROVED, live-ready and trade-enabled
- the stored stress validation is PASS
- the live bar policy permits order execution
- release, ledger and operational integrity checks are healthy

8. Risk management

Quantara's main job after finding an opportunity is deciding how little it can safely risk—or whether to skip it.

Current phase-one account contract

Rule	Configured value	At a $5,000 start
Profit target	8%	$400 gain
Overall drawdown	10%, static; lower of balance or equity	$500 loss floor
Daily drawdown	4%; lower of balance or equity	$200 daily limit
Minimum trading days	3 qualifying days	Each qualifying day ≥ 0.5%
Daily reset	Configured 22:00 UTC	Must be checked for broker/DST alignment

Portfolio operating caps

Cap	Maximum	Approximate dollars at $5,000
One trade	0.20%	$10

Cap	Maximum	Approximate dollars at $5,000
One edge	0.40% open risk	$20
One family	0.80% open risk	$40
One asset	0.95% open risk	$47.50
Whole portfolio	1.70% total open risk	$85
Position count	Up to 5 per edge; 15 per asset	Monetary caps still apply
Minimum-lot inflation	At most 3× intended model risk	Otherwise the trade is skipped

There is intentionally no one-position-per-edge rule. Multiple positions may coexist when they are distinct approved signals and every risk cap remains satisfied. This preserves the portfolio's multi-position design while preventing an edge from consuming unlimited exposure.

How the lot size is formed

The target starts with the edge's relative portfolio weight and its payoff-aware AFML confidence. It is then reduced by drawdown throttles and clipped by every applicable risk ceiling. Finally, the target is converted into an actual broker lot using the symbol's tick value, contract size, currency conversion, minimum lot and lot step. The executor backs down to the largest safe broker step; if even the minimum lot cannot fit, it skips the trade.

As the remaining overall drawdown buffer shrinks, the configured early-soft throttle progressively cuts new risk. The current contract also blocks new trades at or above the profit target. Optional daily-symbol and floating-loss guards exist in the architecture but are disabled in the current account-rules profile; the active 4% daily drawdown and 10% overall drawdown rules remain enforced.

9. Reliability and self-protection

Quantara assumes that files, clocks, broker sessions and processes can fail—and designs explicit states for those failures.

Authoritative broker clock: The dashboard measures several active symbols, locks one account-level offset and publishes an atomic clock contract. The executor normalizes raw broker time before judging tick freshness.

Immutable feature snapshot: Readiness and scoring share the same closed bars, FFD/HMM frames, generation identity and content hashes. Recovery cannot silently swap the evidence underneath a decision.

Per-asset quarantine: If one market's feature chain is incomplete, that asset is prevented from scoring. Healthy markets can continue instead of waiting behind a global repair.

Transactional recovery: A replacement FFD/HMM generation is built in staging, fully validated and activated as a pair. The last verified generation is retained for rollback; active lock files are not archived.

Rolling-window validation: The system checks the entire dependency window, not merely the final row, so an internal gap that poisons a later rolling feature is traced to its origin.

Atomic durable state: Critical state is written atomically with last-good recovery. Closed bars first enter a durable journal before compaction, reducing loss from interruption.

Release identity: Each process records the exact Git commit and a digest of critical runtime files, helping prove which code produced an action.

Truthful observability: Heartbeats separate current-run counters from lifetime state and expose clock health, quarantine, recovery queues, latency, candidate funnel and output parity.

> **FAIL CLOSED** When Quantara cannot prove that required data, clock authority, feature generation, account state or broker risk is valid, the intended behavior is to block, retry within a bounded window, quarantine the affected asset or disarm order sending—not to substitute a guess.

10. A concrete example

This fictional walk-through uses the real XAUUSD/trend_long contract but invented market probabilities and prices.

1. A gold information bar closes after its required number of quote ticks. The bar is new, executable and later than the dashboard's startup cutoff.
2. The exact-readiness layer confirms that the closed-bar frame, frozen FFD context and required HMM regime rows belong to one verified generation.
3. The trend_long candidate logic sees a positive impulse with trend-side confirmation and creates a candidate event.
4. The edge runtime builds the sealed model inputs. Imagine its raw score calibrates to 0.55. The live threshold for XAUUSD/trend_long is 0.52, so the probability gate passes.
5. Payoff-aware AFML produces a fractional sizing strength. The dashboard creates a 1:1 target/stop plan with the edge's 100-information-bar maximum horizon, then writes an approved signal.
6. The executor reads the signal. It checks the 1,000-second age limit, current normalized gold tick, spread, price drift, stop distance, account identity and every open-risk cap.
7. If the largest safe broker-valid lot exists, the order is sent. If the spread is too wide or the minimum lot is too risky, the same model approval ends as a documented rejection instead.
8. After a fill, Quantara monitors the broker stop/target and the vertical bar horizon. The ledger later reconciles the actual broker deals, exit reason and realized result.

> **THE KEY LESSON** Model approval is one checkpoint, not the final command. Research truth, live-data truth, execution quality and portfolio safety must all agree before money is put at risk.

11. What Quantara records

The audit trail is part of the product, not an afterthought.

Runtime score rows: What each edge saw, which features were present, the raw and calibrated probabilities, AFML size and threshold decision.

Signal audit: Every observed candidate and its progression through detection, readiness, scoring and approval.

Approved signal handoff: The append-only CSV/JSONL records consumed by the executor, including the execution plan and identifiers.

Execution audit and rejected signals: Why the executor accepted, retried or blocked a candidate: clock, freshness, spread, drift, risk, lot size, market state or another explicit reason.

Executed trades: The actual lot, fill, stop, target, risk amounts, edge identity, signal identity and broker tickets.

Closed-trade ledger: A reconciliation of expected positions with MT5 deals, including open/closed state, exit reason and unexplained discrepancies.

Status and heartbeat: A live operator view of release identity, broker clock, portfolio exposure, queue depth, latency, quarantines, recovery, counters and last action.

These records make tomorrow's review possible. Speed cannot be judged from an absence of exceptions alone: the useful measurement is the complete funnel from bar close to feature readiness, model scoring, approved handoff, executor plan and order attempt.

12. What Quantara does not do

The system is sophisticated, but its boundaries are as important as its capabilities.

- It does not know the future or guarantee profitable trades.
- It does not trade every bar or every candidate event.
- It does not autonomously retrain models, change thresholds or rewrite strategy logic while live.
- It does not let the dashboard send orders or let the executor rescore models.
- It does not treat a repaired history bar as a real live opportunity.
- It does not assume that a backtest, Monte Carlo pass or two quiet live bars prove production reliability.
- It does not remove broker, VPS, data-quality, slippage, spread, gap or market-regime risk.
- It does not turn a risk cap into a promise that realized loss can never exceed that cap; market gaps and execution can still produce slippage.

How to judge it honestly

A fair evaluation separates four questions. First, does the research have genuine out-of-sample evidence? Second, does the portfolio survive realistic overlap, broker-lot and account-rule stress? Third, does live scoring reproduce the sealed research decision quickly enough? Fourth, do real fills, slippage and realized losses behave close enough to the assumptions? Quantara contains controls for all four, but live evidence must accumulate over time.

> **CURRENT ASSESSMENT BOUNDARY** The repository now contains the unified clock, immutable snapshot, generation-aware recovery and VPS compatibility fixes. The dashboard and executor are running without reported errors so far, but today's full-session logs are still needed to determine whether the previous speed and recovery issues are actually resolved in production.

This guide is an explanation of a software system, not investment advice. Automated trading can lose money, and even a well-designed fail-closed system cannot eliminate market risk or all software defects.

13. Short glossary

The essential terms, without the quant vocabulary barrier.

AFML: Advances in Financial Machine Learning; here, a payoff-aware method for turning calibrated confidence into fractional bet size.

Allocator: The offline portfolio process that chooses the live edge set, thresholds, weights and risk configuration.

Bar: A compact open-high-low-close summary of a sequence of quote ticks.

Calibrated probability: A model score adjusted so it behaves more like a real success probability.

Candidate event: A broad, rules-based situation that asks an edge's model to evaluate an opportunity.

Causal / past-only: Calculated using information that was genuinely available at that time.

Drawdown: The decline from a reference balance/equity level.

Edge: One asset, one economic idea and one direction, with its own runtime and artifacts.

Fail closed: Stop or block when proof is missing, instead of continuing with an assumption.

FFD: Fractional differentiation, a memory-preserving transformation of price history.

HMM: Hidden Markov Model, used to describe changing price regimes as four latent states.

HRP: Hierarchical Risk Parity, a way of allocating relative weight after clustering similar return streams.

Live manifest: The sealed final authority listing selected edges, thresholds, weights, geometry and risk rules.

Meta-label: The training outcome that asks whether a candidate was worth taking under its fixed trade geometry.

MT5: MetaTrader 5, the broker terminal/API that supplies prices, account state and order execution.

Out-of-fold: Predictions produced on data excluded from the model's training fold, used to reduce self-evaluation bias.

Quarantine: A state that prevents one asset from scoring until its dependencies are verified or repaired.

R: One unit of planned trade risk. A 0.05R cost means costs equal five percent of that planned risk unit.

Signal: A candidate that has passed the model and dashboard gates and has an execution-ready plan.

Vertical horizon: The maximum number of information bars a position may remain open under its research contract.

Document basis and scope

This is a reader's guide to the current codebase, not a substitute for the operator runbook.

Repository examined: Wells21/trading_models, branch v2, commit aa7632f783b63d514b0064aa7eca754174e48d66 (“Fix VPS feature recovery compatibility (#16)”), current on 12 August 2026.

Primary evidence reviewed

- Root architecture contracts and SOTD end-to-end runbook.
- Ten-asset and 109-edge registries, representative edge runtimes, event generators, schemas, model-training manifests and deployability reports.
- Allocator ingestion, independence, HRP, broker minimum-lot, rolling challenge, final walkthrough and Monte Carlo stress outputs.
- The 74-edge live manifest, active account rules, live bar policy and environment/manifest validation reports.
- Dashboard, executor, broker-clock, immutable snapshot/readiness, feature continuation/recovery, state integrity, observability and closed-trade ledger code.
- Production hardening and live-streaming validation tests, plus the merged recovery-hotfix history.

Scope notes

- The separate local multi-model strategy research project is not part of Quantara and is intentionally excluded.
- Repository outputs, old backups and forensic bundles were treated as historical evidence; current v2 code, live manifest and active rules were treated as runtime authority.
- Private VPS overrides, credentials and today's still-running logs are not reproduced in this shareable document.
- Current numerical settings are a dated snapshot and may change when a new account, broker, manifest or validated portfolio is deployed.

> **A SIMPLE CLOSING DESCRIPTION** Quantara is a contract-driven automated trading platform: it researches specialist market ideas, proves they can be reproduced live, combines them into a risk-balanced portfolio, observes MT5 markets, approves only calibrated opportunities, and executes only when a separate broker-and-risk gate agrees.